Reliability Engineering and System Safety 259 (2025) 110950

Contents lists available at ScienceDirect

Reliability Engineering and System Safety

journal homepage: www.elsevier.com/locate/ress

Cascading failure analysis of interdependent water-power networks based on functional coupling

Yang Li, Mingyuan Zhang *

Department of Construction Management, Dalian University of Technology, Dalian, Liaoning 116024, China

ARTICLE INFO

Keywords:
Water supply network
Electric power network
Cascading failure
Functional coupling
Earthquake

ABSTRACT

Due to the increasing interdependence and interconnection, the water supply network (WSN) and electric power network (EPN) face a higher risk of cascading failures. Existing studies mainly focus on the cascading failures of the single network but rarely on the interdependent water-power networks (IWPN) under earthquakes. Therefore, combined with the physical operation characteristics, this paper proposes a cascading failure analysis method for interdependent water-power networks based on functional coupling. First, we define the functional coupling relationships between the IWPN and establish a topology model of the IWPN. Subsequently, the joint probability and functional coupling strength are introduced to determine the failure probability of coupled components in the WSN and EPN. The initial failure components are determined by a random method. Then, the node load function and line capacity function are introduced as the judgment conditions of cascading failure of the WSN and EPN, respectively. The cascading failure transmission process of the WSN and EPN is further conducted based on the dynamical flow model. Further, a calculation method for the functional loss of the WSN and EPN is proposed. Finally, the proposed methodology is applied to the coupling WSN of a certain city and IEEE118 node network. The results show that cascading failures in the IWPN spread wider than a single network and cause more serious functional losses. The findings of this work would have important implications for formulating disaster prevention and mitigation measures and seismic performance improvement strategies for interdependent infrastructure networks.

1. Introduction

The WSN and EPN are typical representatives of the urban infrastructure networks, and play a vital role in daily production and life and post-disaster recovery and reconstruction. The earthquake damage data over the years show that the damage to the WSN and EPN will inevitably lead to extremely serious consequences. For instance, after the Hokkaido earthquake in Japan on September 6, 2018, power facilities were damaged, resulting in an almost nationwide blackout that lasted for two days. Eventually, the road transportation network was paralyzed, the water supply network was temporarily closed, and the airport was suspended. The cascading failures of the infrastructure networks under earthquakes caused losses of approximately $2.759 billion. On February 6, 2023, an M_w 7.8 earthquake hit Turkey, followed by another M_w 7.5 earthquake after nine hours. It caused severe damage to infrastructure networks, including transportation networks, power distribution networks, and water supply networks, thereby seriously impeding post-earthquake emergency response and reconstruction work. Ultimately, the number of casualties was approximately 60,000 and the combined direct economic losses incurred by Turkey and Syria were estimated at $100 billion [1]. Traditional earthquake damage research on the WSN and EPN usually only conducts reliability or resilience analysis on a single system, ignoring the interdependent relationships between the two networks. However, the WSN and EPN are highly correlated and complexly coupled. For example, water plants require an electricity supply to maintain normal operation, and electric pumps lift water to higher ground or floors, which also requires an electricity supply; cooling water systems also play an important role in the operation of power plants or nuclear power plants. Due to increasingly complicated interdependences and dynamic network load distribution in the IWPN, a small localized disturbance could trigger domino effects, causing large-scale system failure propagation. This recursive circumstance is called cascading failures, which can be disastrous, and even affect millions of people with inestimable costs [2,3]. Therefore, the interdependent relationships between the WSN and EPN cannot be ignored. In-depth research on the cascading failure process of the IWPN under

* Corresponding author.
E-mail addresses: ly2021dg@mail.dlut.edu.cn (Y. Li), myzhang@dlut.edu.cn (M. Zhang).

https://doi.org/10.1016/j.ress.2025.110950
Received 15 January 2024; Received in revised form 28 January 2025; Accepted 19 February 2025
Available online 20 February 2025
0951-8320/

earthquake damage is of great significance to improving the disaster resilience of the urban infrastructure networks.

To explore the above catastrophic phenomena in IWPN, it is first necessary to explore the method of establishing interdependent relationships between the WSN and EPN. According to existing research, the research on the interdependence of infrastructure networks is relatively mature. Rinaldi et al explored the challenges and complexities of infrastructure interdependencies from six important dimensions, including type of interdependencies, infrastructure environment, coupling and response behavior, type of failure, infrastructure characteristics, and state of operations [4]. This work argued that only through reasonable modeling and simulation methods can we fully understand and analyze the interdependencies of infrastructure networks. Afterward, a large number of modeling and simulation methods targeting the interdependencies between infrastructure networks emerged in the academic community. Through literature reviews, Ouyang [5] summarized the modeling methods of interdependencies, which can be roughly divided into the following categories: empirical approaches (such as empirically-based risk analyses [6]), systems dynamic-based approaches [7], economic theory-based approaches (such as the Leontief input-output model [8,9]), agent-based approaches (such as the Aspen [10] and SMART II [11] projects led by Sandia National Laboratory and Argonne National Laboratory in the United States), and network-based approaches (such as topology-based methods [12] and flowed-based methods [13]). In recent years, new methods for modeling interdependencies have emerged one after another [14,15]. Sharma et al [16] presented a novel classification for infrastructure interdependencies that is consistent with their mathematical modeling based on their ontological and epistemological dimensions. The dimension of ontology classifies the interdependencies based on when and how certain interdependencies exist. The dimension of epistemology classifies the interdependencies in consistency with the mathematical models used to capture them. The above models and methods characterize the interdependencies of infrastructure networks from different perspectives and deepen our understanding of the mechanisms of interdependency. In fact, the intuitive representation and computational convenience of network graphs make network-based approaches one of the mainstream methods in the field of infrastructure system research. In a network model, nodes represent infrastructure system facilities and structures, while connections represent the interactive relationships between infrastructure system facilities and structures [17]. Network-based approaches emphasize the topological structure and flow of infrastructure networks, use network connections to intuitively represent various interdependencies, and can use mature graph theory and complex network theory to develop quantitative analysis tools. Therefore, a large number of interdependent infrastructure modeling studies have adopted network-based approaches.

Key technological, economic, and regulatory changes have greatly altered the relationships between infrastructure networks. Intelligent technologies have led to a significant increase in the interconnectivity and complexity of infrastructure with generally greater centralization of control. When disasters occur, such interdependencies will trigger cascading effects, greatly increasing the impact of disasters on infrastructure networks. Over the past decade, research on cascading failures of interdependent infrastructure networks has increased extensively, aiming to analyze the disaster propagation mechanism in detail and take effective measures to block it and reduce losses. Cascading failure is a potentially destructive process that causes local or small faults to spread within and between systems through the interconnected relationships between system components, and may cause serious system losses and affect the integrity of the infrastructure system [18]. Compared with other phenomena of infrastructure system damage, cascading failure emphasizes the continuous dynamic propagation process of failures and effectively represents the actual performance degradation of infrastructure networks. For instance, Rahnamay-Naeini and Hayat [19] proposed a novel interdependent Markov-chain framework that enabled capturing interdependencies between two critical infrastructures with the ultimate goal of predicting their resilience to cascading failures and characterizing the effects of interdependencies on system reliability. Wu et al [20] combined statistical methods to build a cascading failure propagation model based on discrete-time dynamical systems. The failure probability of a certain unit was determined by the state of the unit at the last moment. The conditional probability of cascading failures between units was used to express the cascading relationship. Furthermore, the propagation rule of failure probability should satisfy the probability measure. Obviously, these studies cannot simulate the network flow changes of real interdependent infrastructure networks during the cascading failure process. The reason is that it did not consider the physical operational characteristics of real infrastructure networks, such as hydraulic models of water supply systems, power flow models of power systems, traffic flow models of transportation systems, and so on. For real infrastructure systems, considering the changes in network flows during cascading failures can better reflect the functional loss of infrastructure systems. Subsequently, Guidotti et al [21] presented a probabilistic approach to modeling network resilience that considered dependency on other networks and that incorporates both physical damage and network functionality to estimate system recovery as a function of time. Xiao et al [22] presented a framework and method for assessing the seismic resilience of interdependent lifeline networks. This framework considers the bidirectional interdependence of lifeline networks and introduces the concept of limited recovery to exemplify the difference in performance between pre and post-disaster. The performance or functionality of the above studies considers the network flow of infrastructure systems. Moreover, the above studies also consider the direct physical damage of earthquakes to network components and the initial cascading effects between networks caused by interdependencies. However, these cascading effects are instantaneous. In infrastructure systems, due to the capacity or threshold limitations of component network flows, such cascading effects will continue, causing more serious functional losses.

As for the WSN and EPN, they are highly correlated and complexly coupled. In existing research, interdependency modeling of the water-power network is also mostly done using network-based approaches [23,24]. When disasters occur, the damage to the water or power network is more severe due to their interdependence, mainly because of the cascading effect. Despite extensive studies on the cascading failures in interdependent infrastructure networks, there are few studies on cascading failures for the IWPN. Zhang et al developed a more realistic cascading failure model to explore the vulnerability of interdependent power-water networks [25]. Moreover, Lee et al [26] proposed a Bayesian network-based seismic damage estimation method for power and potable water supply system. However, the above approaches ignore the failure modes and failure mechanisms of system structural components under specific disasters. Furthermore, the physical operation characteristics of the WSN and EPN are also ignored, which could largely overestimate the disaster resistance capacity of the network. In response to these shortcomings, Pournaras et al [24] conducted cascading failures in interconnected power-to-water networks. Based on power and water flow dynamic simulation, the impact of power cascading failures on shortages of water supply was revealed. Although the physical operation characteristics of the system are emphasized, the above research only analyzes the impact of the EPN failure on the WSN, that is, one-way cascading failure propagation. In fact, the physical connection and functional dependence between the WSN and EPN are bidirectional. Specifically, the water plant or reservoir provides cooling water to the power plant. Therefore, in this way, it is possible to analyze whether failures of interconnected components in a WSN or EPN may be transmitted through components to disrupt the operation of the EPN or WSN.

On the other hand, in the context of frequent earthquakes, great achievements on seismic damage of a single WSN or EPN have been made [27,28]. However, there are limitations in the cascading failures of the IWPN under earthquake scenarios. Given the above research gap, this paper takes the IWPN as the research object and proposes a cascading failure analysis method based on functional coupling. Firstly, this paper defines the functional coupling relationships between the WSN and EPN from the perspective of the functional supply and demand relationship of system components and establishes the topology model of the IWPN based on graph theory. Later, the seismic fragility analysis [29] is employed to determine the probability of different damage states of components, and the failure probability of coupled components caused by earthquake damage and dependency relationships is calculated by the joint probability method, where the functional coupling strength is defined to represent the possibility of failure transmission. Next, the initial failure components are determined by a random method. More importantly, the above initial failure components are used as the staged starting point of cascading failure, and the node load function and line capacity function are introduced as the judgment conditions for cascading failure of water network nodes and power network lines. The subsequent failure transmission process in the WSN and EPN is further simulated based on dynamic network flow. On this basis, the supply loss rate and link loss rate are introduced as functional evaluation indicators to analyze the changes in the overall functional loss of the system caused by this failure transmission. In order to verify the applicability and feasibility of this method, this paper combines case studies to compare the functional losses of the WSN and EPN with and without functional coupling relationships under different earthquake scenarios. In addition, since the functional coupling strength and tolerance threshold are not a definite value, this paper implements a sensitivity analysis and obtains the influence of the above parameters on the functional loss of the WSN and EPN under different earthquake scenarios. Overall, the main contribution of this paper is to apply physical operation characteristics and functional coupling relationships to study the cascading effects of the IWPN under earthquakes, which can efficiently improve the accuracy of loss estimation of bidirectional interdependent infrastructure networks.

The remainder of this paper is organized as follows. In Section 2, we define the functional coupling relationships between the IWPN, and the IWPN topology model is established. Based on the hydraulic model and DC power flow model, the cascading failure analysis of the WSN and EPN is conducted by considering the tolerance capacity of water network nodes and power network lines. Further, a calculation method for functional loss of the WSN and EPN is proposed. Section 3 presents a case study where cascading failure simulation results and sensitivity analysis results are provided. Section 4 discusses the research results. Finally, Section 5 draws some concluding remarks and points out future research directions.

2. Methodology

There are various types and quantities of unit components in urban WSN and EPN, along with a wide range of geographical distribution. Under the action of an earthquake, one component of the WSN or EPN (such as the water plant, pumping station, power plant, substation, etc.) is damaged, which could cause damage to other components in the system. In addition, due to the interdependence between the WSN and EPN, damage may affect the operation of the other network. Therefore, this paper proposes a method for analyzing the cascading effect of the IWPN under earthquakes. The flowchart of specific idea is shown in Fig. 1, including three steps: (1) establishing a topological structure model of the IWPN; (2) analyzing the cascading failure process of the IWPN; (3) calculating the post-earthquake functional losses of the WSN and EPN.

2.1. Topology Structure of the IWPN

To represent mathematically a generic infrastructure, or multiple systems of infrastructure, it is possible to model them as networks [30]. Based on the fundamental concepts of graph theory, a generic network is defined by the set of nodes or vertices and the set of links or edges that connect the nodes. In this paper, the WSN is displayed as $G^W(V^W, E^W)$. $V^W = (v_1^W, v_2^W, \cdots, v_{N1}^W)$ is the set of $N1$ nodes, which contain water plants, water towers, and users. $E^W = (e_1^W, e_2^W, \cdots, e_{N2}^W)$ is the set of $N2$ links, which denote pumping stations and pipelines of the WSN. The EPN is expressed as $G^H(V^H, E^H)$. $V^H = (v_1^H, v_2^H, \cdots, v_{M1}^H)$ is the set of $M1$ nodes, which represent power plants, substations, and users. $E^H = (e_1^H, e_2^H, \cdots, e_{M2}^H)$ is the set of $M2$ links, which show transmission and distribution lines of the EPN (Table 1).

From the perspective of functional supply and demand, the interdependent relationships between the WSN and EPN is established in this paper. Specifically, there is a certain connection between a component i of the WSN and a component j of the EPN, and the status of component i may affect component j [4]. This paper analyzes the functional coupling relationships between the WSN and EPN, which is consistent with the concept of functional interdependence proposed by Zhang and Peeta [31], namely, the functional realization of a network requires the input of the associated network. Generally speaking, two networks with functional coupling relationships are first topologically dependent on each other. When one network is interrupted, the cascading failure will occur in itself or the associated network due to the function transferability. For instance, the operation of the pumping station in the WSN depends on an electric power supply. When the EPN is damaged, the pumping station could fail to function. Table 1 shows the functional coupling relationships between the WSN and EPN. As for the EPN, the power plant is regarded as the functional coupled component, because its normal operation requires the cooling water from water plants or pools. And the function transfer direction is from a water plant or pool to a power plant, that is, when the water plant or pool is damaged, the function of the power plant will be weakened or failed. In terms of the WSN, the functional coupled component is the water plant and pumping station. Specifically, the stable electric power supply from the substation can ensure the normal operation of the water plant and pumping station, and the function transfer direction is from a substation to a water plant or pumping station.

A modeling framework of multiple infrastructure networks [31] is adopted to establish the functional coupling relationships between the IWPN. Fig. 2 shows a schematic diagram of the IWPN topology structure.

2.2. Cascading Failure Analysis of the IWPN

2.2.1. Initial Trigger Judgment of Cascading Failures

The degree of damage to each component of the WSN and EPN is closely associated with seismic parameters. The earthquake intensity table of the “Seismic ground motion parameters zonation map of China” (GB18306–2015) [32] is adopted to determine the seismic shaking parameters. Subsequently, the Hazus Earthquake Model Technical Manual (Hazus 5.1) [29] released by FEMA is used to obtain the fragility curve of each component of the WSN and EPN. Through the seismic fragility curve, the probability of different damage states of each component under different earthquake scenarios can be determined. In order to simplify the calculation, the damage states of each component in the WSN and EPN are divided into two types, namely, normal functional status and failure functional status. According to the “Guideline for evaluation of seismic resilience assessment of urban engineering systems” (RISN-TG041–2022) [33], this paper defines that components reaching a moderate damage level cannot operate normally, which indicates functional failure. A random number between 0 and 1 is used to judge the functional failure of each component. When the functional failure probability of the component exceeds a random number, the component fails.

2.2.2. Functional Failure Analysis of Coupled Components

In this paper, the joint probability is introduced to determine the

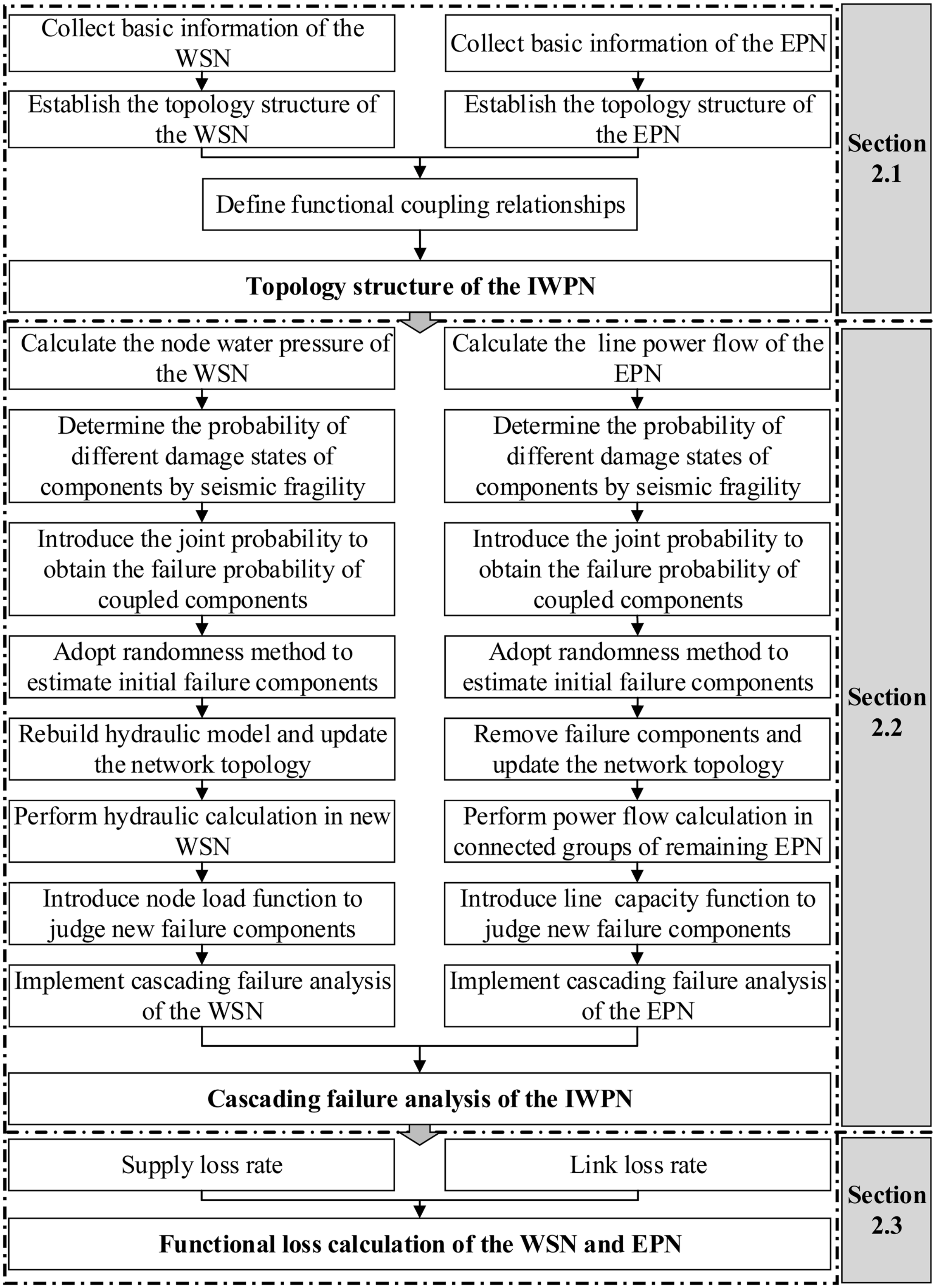


Fig. 1. Flowchart of cascading failure analysis method of the IWPN.

Table 1
The functional coupling relationships between the WSN and EPN.

Infrastructure network	Functional coupled component	Functional coupling relationship	Function transfer direction
EPN	Power plant	The operation of the power plant requires cooling water from water plants or pools	Water plant/ pool→ Power plant
WSN	Water plant	The operation of the water plant requires electric power supply from the substation	Substation→ Water plant
	Pumping station	The operation of the pumping station requires electric power supply from the substation	Substation→ Pumping station

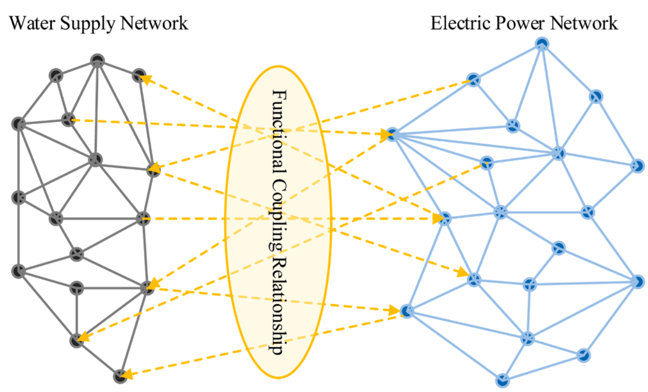


Fig. 2. Schematic diagram of the IWPN topology.

failure probability of coupled components. Moreover, the functional coupling strength is proposed to analyze the possibility of component functional failure transmission between different networks, which is described in detail as follows.

A. Joint Probability. According to the description in Section 2.1, the functional failure of the coupled component u in the WSN W under the action of earthquakes originates from two situations: (1) the direct damage of the coupled component u is caused by the earthquake, which is recorded as an event W_u^{sei}, and its occurrence probability is expressed as $P(W_u^{sei})$. (2) the functional failure of component v in the EPN H that component u depends on is transmitted to component u, which is recorded as an event W_u^{cas}, and its occurrence probability is denoted as $P(W_u^{cas})$. In terms of the failure of the coupled component u, that is, the event W_u, a joint probability model [34] is introduced to represent the occurrence probability of the event $P(W_u)$, as shown in Eq. (1). Combined with the above analysis, the event W_u can be regarded as the union of the event W_u^{sei} and the event W_u^{cas}.

$$P(W_u) = P\left(W_u^{sei} \bigcup W_u^{cas}\right) \tag{1}$$

And then, Eq. (1) can be further equivalent to:

$$P(W_u) = P\left(W_u^{sei}\right) + P\left(W_u^{cas}\right) - P\left(W_u^{sei} \cap W_u^{cas}\right) \tag{2}$$

The damage events of component u and component v are independent of each other, so Eq. (2) can be converted to:

$$P(W_u) = P\left(W_u^{sei}\right) + P\left(W_u^{cas}\right) - P\left(W_u^{sei}\right) * P\left(W_u^{cas}\right) \tag{3}$$

where the failure probability $P\left(W_u^{sei}\right)$ of component u is determined according to Section 2.2.1. The calculation method of $P\left(W_u^{cas}\right)$ is shown as in Eq. (4).

$$P\left(W_u^{cas}\right) = P\left(W_u^{cas}|H_v\right)P(H_v) \tag{4}$$

In Eq. (4), the cascading failure probability $P\left(W_u^{cas}\right)$ of component u is computed as the product of the failure probability $P(H_v)$ of component v and the conditional probability $P\left(W_u^{cas}|H_v\right)$ of the associated failure of component u under the condition that component v fails. It is indicated that the cascading failure event W_u^{cas} of component u is caused by the failure of component v in the EPN. $P(H_v)$ is the failure probability of component v. If component v has a functional coupling relationship with other components, then $P(H_v)$ is solved according to Eq. (1); otherwise, $P(H_v)$ is the initial functional failure probability of component v under earthquake damage. Referring to existing research [35], the conditional probability $P\left(W_u^{cas}|H_v\right)$ is equivalent to the coupling strength Φ_{uv} between component u and component v, as described in the next section.

B. Functional Coupling Strength. The functional coupling strength Φ_{uv} is defined as the possibility of functional failure transmission between component u and component v, and the transmission direction is from component u to component v. The value of Φ_{uv} is assumed as follows.

$$\Phi_{uv} = \begin{cases} 0 & \textit{none coupling} \\ 0 \sim 1 & \textit{weak coulping} \\ 1 & \textit{strong coupling} \end{cases} \tag{5}$$

when $\Phi_{uv} = 0$, it means that there is no coupling relationship between component u and component v, namely, when the function of component u fails, the failure will not be passed to component v. When $0 < \Phi_{uv} < 1$, it is indicated that there is coupling relationship between component u and component v. Specifically, when Φ_{uv} is close to 0, it shows that the coupling relationship is relatively loose, and the failure effect of component u is less likely to be transmitted to component v. When Φ_{uv} is close to 1, it suggests that the stronger the coupling relationship, the easier it is for the failure effect of component u to be transmitted to component v, resulting in functional failure of component v. When Φ_{uv} is equal to 1, the two components are strongly coupled, that is, the functional failure of component u will inevitably lead to the functional failure of component v.

2.2.3. Cascading Failure Analysis of the WSN

According to the above descriptions, the initial functional failure probability of components, including reservoirs, water towers, and pumping stations, is obtained. In addition, it is assumed that pipeline damage under earthquake action is a random independent event and obeys Poisson distribution along the length of the pipeline [36]. Therefore, the probability of pipeline damage from earthquakes is described in Eq. (6).

$$P_{b,l} = 1 - e^{(-RR_l \cdot L_l)} \tag{6}$$

where $P_{b,l}$ is the probability of pipeline l being damaged; L_l is the length of pipeline l; RR_l denotes the average seismic damage rate of pipeline l, which is calculated using the fitting method of existing seismic motion data and damage data of pipelines [37], as shown in Eq. (7). D is the diameter of pipeline l.

$$\ln(RR_l) = \begin{cases} 1.84 \cdot \ln(PGV) - 9.40 & \textit{Ductile iron pipe}(D < 600mm) \\ 1.41 \cdot \ln(PGV) - 8.19 & \textit{Steel pipe}(\textit{riveted joints}, D \geq 600mm) \end{cases} \tag{7}$$

Two damage states of the pipeline include breakage and leakage. Surveys showed that 80% of the damage status of pipelines after the earthquake was leakage, and 20% was breakage [29]. Referring to existing research, it is assumed that the leakage probability of the pipeline is 5 times the breakage probability, and a random number between 0 and 1 is used to judge the damage state of the pipeline. Under the action of earthquakes, a large number of pipelines are damaged, which results in the system being in a low-pressure operating state. In such a circumstance, the water demand of some nodes cannot be fully satisfied. It is widely known that the actual water distribution volume of a node is closely related to the node pressure. According to the node flow continuity equation, if the node pressure is reduced to meet the fixed water demand, the hydraulic adjustment calculation result will show node negative pressure, which is not consistent with the actual situation. Therefore, Wagner's model [37] is adopted to calculate the actual water distribution volume for the post-earthquake hydraulic analysis of the WSN, and the calculation method is given in Eq. (8). In addition, the post-earthquake hydraulic model of the WSN [38] is utilized to perform hydraulic adjustment calculation. The hydraulic model of the WSN is established according to the damage status of the pipeline. Specifically. when the pipeline is broken, two virtual pools are added at the pipeline disconnection position. When the damaged state of the pipeline is in leakage, a diffuser is added at the pipeline damaged location. Adding the hydraulic model into the pre-earthquake and modifying the pre-earthquake topology to form the post-earthquake topological structure of the WSN. Further, the epanet2.dll function in Epanet is called to perform the hydraulic calculation of the WSN.

$$Q_k^{act} = \begin{cases} 0 & P_k \le P_k^{min} \\ Q_k^{req}\sqrt{\dfrac{P_k - P_k^{min}}{P_k^{ser} - P_k^{min}}} & P_k^{min} < P_k < P_k^{ser} \\ Q_k^{req} & P_k^{ser} \le P_k \end{cases} \tag{8}$$

where Q_k^{act} and Q_k^{req} denote the actual distribution volume and demand volume of node k, respectively. P_k is the water pressure of node k; P_k^{min} and P_k^{ser} represents the minimum and the service water pressure of node k, respectively.

After the earthquake, initial damage occurs to each component, including the pool, water tower, pumping station, and pipeline. In this circumstance, the service water pressure of most nodes cannot be satisfied, indicating that these nodes lose service functions. It is obvious that too low nodal pressure will cause flow interruption and water outage, so a specific minimum water pressure range (minimum bearing capacity) should be maintained to ensure a stable water supply. Therefore, node failure is regarded as the inability of a component in the WSN to provide sufficient flow or pressure to meet demand. In this paper, the minimum water pressure of a node is defined as the lowest value of the nodal pressure designed for the WSN, that is, the minimum bearing capacity of the node is proportional to the initial pressure [39]. The calculation method is presented as follows:

$$P_k^{min} = (1 - \alpha_k)P_k^{ser} \tag{9}$$

where α_k is the tolerance value of node k, and the range is 0~1. The larger the value of α_k is, the lower the probability of node failure.

To sum up, the cascading failure process of the WSN is summarized as follows:

Step 1: Obtain the basic information of the WSN, and calculate the initial water pressure of each node.
Step 2: Adopt the seismic fragility to obtain the probability of different damage states of components and estimate the functional failure of components using randomness method.
Step 3: Introduce the joint probability and functional coupling strength to obtain failure probability of coupled components and estimate the functional failure of the coupled components using randomness method.
Step 4: Build a post-earthquake hydraulic model when the pipeline leaks or breaks. If a node fails, it is considered that the upstream and downstream pipelines connected to the failed node are broken, and a hydraulic model of the broken pipeline is constructed similarly.
Step 5: Call the epanet2.dll package in the Epanet toolkit to perform hydraulic adjustment to obtain the node pressure. Then the Wagner's model is used to conduct the post-earthquake hydraulic calculation to obtain the water distribution volume.
Step 6: Introduce the node load function to judge the new failure components. If the node pressure is less than the minimum bearing capacity range, the node is regarded as a new failed node and returns to step 4; otherwise, the cascading failure process of the WSN ends.

2.2.4. Cascading Failure Analysis of the EPN

Power flow is the main state variable that describes the dynamic process of electric energy transmission and distribution. The large-scale change of power flow is reflected as the cascading failure process in the EPN. It is an efficient prior research method to establish a large-scale power outage accident model through power flow equations and study the causes, propagation speed, and scope of cascading failures by examining power flow dynamics. The DC power flow equation only considers the distribution of active power flow in the EPN, and has the advantages of fast calculation and no convergence problems. The main purpose of this paper is to analyze the occurrence and propagation process of cascading failures in the EPN after earthquakes and obtain the changes in load node power. Therefore, the DC power flow equation is used to calculate the functional loss of the EPN after earthquakes.

In the DC power flow model, the following assumptions about the actual EPN are made. (1) The node voltage usually runs near the rated voltage, that is, $U_i \approx U_j \approx 1$; (2) The phase angle difference between the two ends of the branch is very small, namely, $\theta_{ij} \approx 0$; (3) The impedance ratio of the ultra-high voltage power network line is very small, that is, $r_{ij} \approx 0$.

Ignoring the parallel branches in the EPN, the active power flow equation of the branch l_{ij} (DC power flow equation) can be approximated as:

$$\begin{cases} P_{ij} = -b'_{ij}(\theta_i - \theta_j) = \dfrac{(\theta_i - \theta_j)}{x_{ij}} \\ Q_{ij} = 0 \end{cases} \tag{10}$$

where P_{ij} and Q_{ij} are the active power and reactive power of branch l_{ij}, respectively. $b'_{ij} = -1/x_{ij}$, x_{ij} is the branch reactance.

According to Kirchhoff's law, the current balance equation of the node i is computed as:

$$P_i^{SP} = \sum_{j \in i, j \ne i} P_{ij} = \sum_{j \in i, j \ne i} \frac{\theta_i - \theta_j}{x_{ij}}, i = 1, 2, \cdots, n \tag{11}$$

where P_i^{SP} is the injected active power of node i; n is the number of nodes. Eq. (11) is further converted into matrix form, namely, the actually used DC power flow equation is shown as follows.

$$P^{SP} = B\theta \tag{12}$$

where P^{SP} represents the active injection power matrix; B is the $n \times n$ order nodal admittance matrix, and its diagonal elements and non-diagonal elements are expressed as follows, respectively.

$$\begin{cases} B_{ii} = \sum_{j \in i, j \ne i} \dfrac{1}{x_{ij}} \\ B_{ij} = -\dfrac{1}{x_{ij}} \end{cases} \tag{13}$$

The cascading failure process of the EPN is closely related to the post-earthquake damage status of each component. The main components, power plants, substations, loads, and transmission and distribution lines, constitute the EPN. Therefore, in order to analyze the cascading failure process of the EPN, it is first necessary to determine the initial functional failure of each component in the EPN, and the calculation method is described in Section 2.2.1. In the EPN, the successive failures of transmission lines are the main driving factors for the cascading failure propagation. Thus, transmission lines are selected as the research components of cascading failure in this paper. Each transmission line in the EPN is equipped with a relay. When the current in the transmission line exceeds the capacity limit and continues for a long time, the overcurrent relay will protect the transmission line and cause it to trip. Therefore, it is assumed that the transmission line capacity C_i is proportional to the initial power flow $f_i(0)$, and the formula is as follows:

$$C_i = (1+\beta_i)f_i(0) \tag{14}$$

where β_i $(0<\beta_i\leq 1)$ represents the tolerance parameter of transmission line *i*. When the power flow of a transmission line exceeds the line capacity, the transmission line is disconnected.

In summary, the detailed cascading failure process of the EPN is listed as follows.

Step 1: Obtain the basic information of the EPN, solve the DC power flow model, and calculate the initial power flow $f_i(0)$ of the transmission line *i*.
Step 2: Adopt the seismic fragility to obtain the failure probability of each component and use randomness method to estimate the functional failure of each component.
Step 3: Introduce the joint probability and functional coupling strength to obtain failure probability of coupled components and use randomness method to estimate the functional failure of coupled components.
Step 4: Remove malfunctioning components and update the network topology.
Step 5: Determine the connectivity of the remaining EPN. If the connected group contains the source node, go to the next step; otherwise, the cascading failure process of the EPN ends.
Step 6: Calculate the transmission flow of the line in the current connected group according to the DC power flow equation.
Step 7: Introduce line capacity function to judge new failure components. If the transmission traffic of the line exceeds the transmission capacity, the line is regarded as a new failed component and returns to step 4; otherwise, the cascading failure process of the EPN ends.

Combined with the descriptions in Section 2.2.1-2.2.4, the cascading failure process of the IWPN is concluded as: Firstly, the earthquake causes damage to the WSN and EPN components, and the seismic fragility is used to determine the probability of different damage states of the components. As for the uncoupled component, the functional failure probability is defined as the probability of reaching or exceeding the moderate damage state; while the functional failure probability of the coupled component is the probability of functional damage caused by seismic damage and cascading effects, as calculated in Eqs. (1)-(5). Specifically, when the function of a water plant or pumping station coupled to a substation fails, the joint probability and functional coupling strength is used to determine the probability of functional failure of the water plant or pumping station. When the function of a power plant coupled with a water plant or pool fails, the functional failure probability of the power plant is determined in the same way. And further, the functional failure of any component is determined by randomness method. Subsequently, the water distribution volume of each node in the WSN and the load of each node in the EPN are calculated based on the hydraulic model and DC power flow model. Through Eq. (9) and (14), the new failure component can be yielded to update the topology structure of the WSN and EPN, respectively. By repeating the above failure determination process until there are no new functional failure components in the IWPN, the cascading failure is terminated.

2.3. Functional Loss Calculation of the WSN and EPN

Water and power supply services originate from the WSN and EPN to ensure people's production and life. Thus, from the perspective of supply and demand, the supply loss rateF_{loss}^{DSR} and link loss rate F_{loss}^{LSR} are proposed to represent the functional loss of any network. The former refers to the ratio between the actual supply of node and the demand; the latter is the ratio of the number of links in a failure state after cascading failure to the initial number of links. Due to the random nature of earthquake damage, the Monte Carlo method is used for simulation calculations. The average value of supply loss rate and link loss rate of the WSN and EPN are taken as the final functional loss calculation results. Therefore, the calculation formula is as follows.

$$DSR = \frac{\sum_{i=1}^{m} Q_i^k}{\sum_{i=1}^{m} Q_i} \tag{15}$$

$$LSR = \frac{L^k}{n} \tag{16}$$

$$F_{loss}^{DSR} = 1 - \frac{1}{N}\sum_{k=1}^{N} DSR \tag{17}$$

$$F_{loss}^{LSR} = 1 - \frac{1}{N}\sum_{k=1}^{N} LSR \tag{18}$$

where Q_i^k and Q_i are the actual supply and demand of the *ith* node in any network in the *kth* simulation, respectively; L^k is the number of surviving links in any network in the *kth* simulation; *DSR*and *LSR* are the demand satisfaction rate and the connectivity rate of any network, respectively. *m* and *n* are the number of nodes and links in any network, respectively; *N* is the number of Monte Carlo simulations.

3. Case Study

3.1. Basic Information

To show the applicability and feasibility of the proposed methodology, a city's water supply network and the IEEE 118 node network were selected to conduct a case study. Functional loss results for the WSN and EPN with and without coupling relationships were calculated under four earthquake scenarios, respectively. Meanwhile, a sensitivity analysis of the functional loss of the WSN and EPN with and without coupling relationships to the functional coupling strength and tolerance parameter was performed. According to the "The Chinese seismic intensity scale" (GB/T 17,742–2020) [40], the seismic intensity in this paper contains VI, VII, VIII, and IX, and the corresponding peak ground accelerations are 0.06, 0.15, 0.3, and 0.6g, respectively. Moreover, the probability of different damage states of each component in the WSN and EPN under a certain earthquake scenario can be determined based on the Hazus Earthquake Model Technical Manual (Hazus 5.1). In addition, the number of Monte Carlo simulations is set to 1000, and the minimum and service water pressure of node *k* are $P_k^{min} = 0m$ and $P_k^{ser} = 10m$, respectively. The tolerance values in the WSN and EPN are assumed as 0.5 and 0.5, respectively.

3.2. Topology Model of the IWPN

Fig. 3 illustrates the topological structure of a city's WSN coupled with the IEEE 118-node network. The WSN contains 4 water source nodes (Junction 50, 51, 52, 53) with total water heads of 76.95, 74.31, 83.58, and 81.98m, respectively, 49 user demand nodes, and 78 pipelines. The Hazen-Williams roughness coefficient of the first pipe section is 80, that of the 44th, 45th, 46th, and 49th pipe sections is 90, and that of the remaining pipe sections is 120. In addition, it is assumed that for

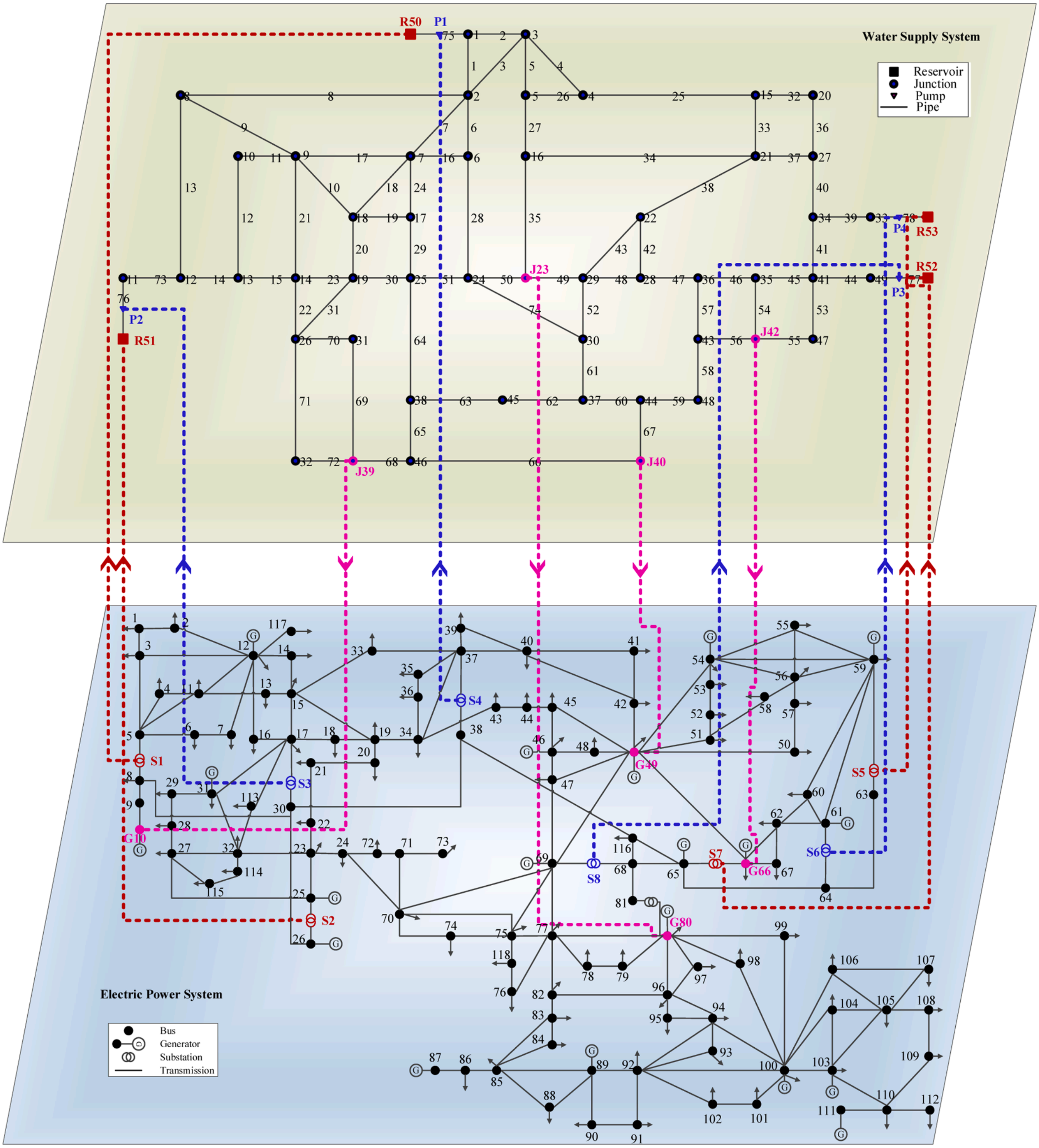


Fig. 3. Topological Model of the IWPN.

pipelines with a diameter greater than 600 mm, the pipe material is steel pipe (riveted joint); for pipelines with a diameter less than 600 mm, the pipe material is ductile iron pipe. More information about the WSN can be found in the literature [38]. The IEEE 118-node network includes 118 buses, 19 generators, 35 synchronous capacitors, 64 load nodes, and 186 lines (including 9 transformers). The detailed parameters come from the case information in the MATPOWER software. Nine transformers are assumed as the substations in this paper, and the information is listed in Table 2.

As shown in Fig. 3, the IWPN contains 12 pairs of functional coupling elements. In the WSN, there are functional coupling relationships between reservoirs or pumping stations and substations, that is, the substation provides power supply to the reservoir or pumping station. Assuming that a coupling relationship between the substation and reservoir is weak. The functional coupling strength Φ_{sr} is set to 0.5, and the function transfer direction is from the substation to the reservoir. In other words, when the substation fails, there is a 50% possibility that the failure will be transmitted to the reservoir, causing the reservoir to fail.

Table 2
The substation information.

From bus	To bus	Substation
8	5	S1
26	25	S2
30	17	S3
38	37	S4
63	59	S5
64	61	S6
65	66	S7
68	69	S8
81	80	S9

In this case, the functional coupling node pairs composed of the substation and the reservoir are (S1, R50), (S2, R51), (S7, R52), and (S5, R53). Subsequently, it is defined that there is a strong coupling relationship between the substation and the pumping station, and the functional coupling strength Φ_{sp} is set to 1. The function transfer direction is from the substation to the pumping station. The functional coupling node pairs consisting of the substation and the pumping station are (S4, P1), (S3, P2), (S8, P3), and (S6, P4). In the EPN, there is a functional coupling relationship between the power plant and the water user, that is, the WSN needs to provide cooling water for the power plant. It is assumed that there is a weak coupling relationship between the power plant and the water user. The functional coupling strength Φ_{jg} is 0.75, and the function transfer direction is from the water user to the power plant. Similarly, there is a 75% possibility of passing the failure of the water user to the power plant. The functional coupling node pairs composed of the water user and the power plant are (J39, G10), (J40, G49), (J42, G66), and (J23, G80).

3.3. Cascading Failure Simulation of the IWPN

3.3.1. Cascading Failure Results

Based on the description in Section 2.2, the cascading failure simulation of the IWPN includes two modules, the coupled WSN simulation module and the coupled EPN simulation module. The former adopted Wagner's model and the epanet2.dll package to perform hydraulic adjustments to calculate the node water volume. Further, the node load function was introduced to reflect the cascading failure propagation effect. The latter utilized the DC power flow model to calculate the

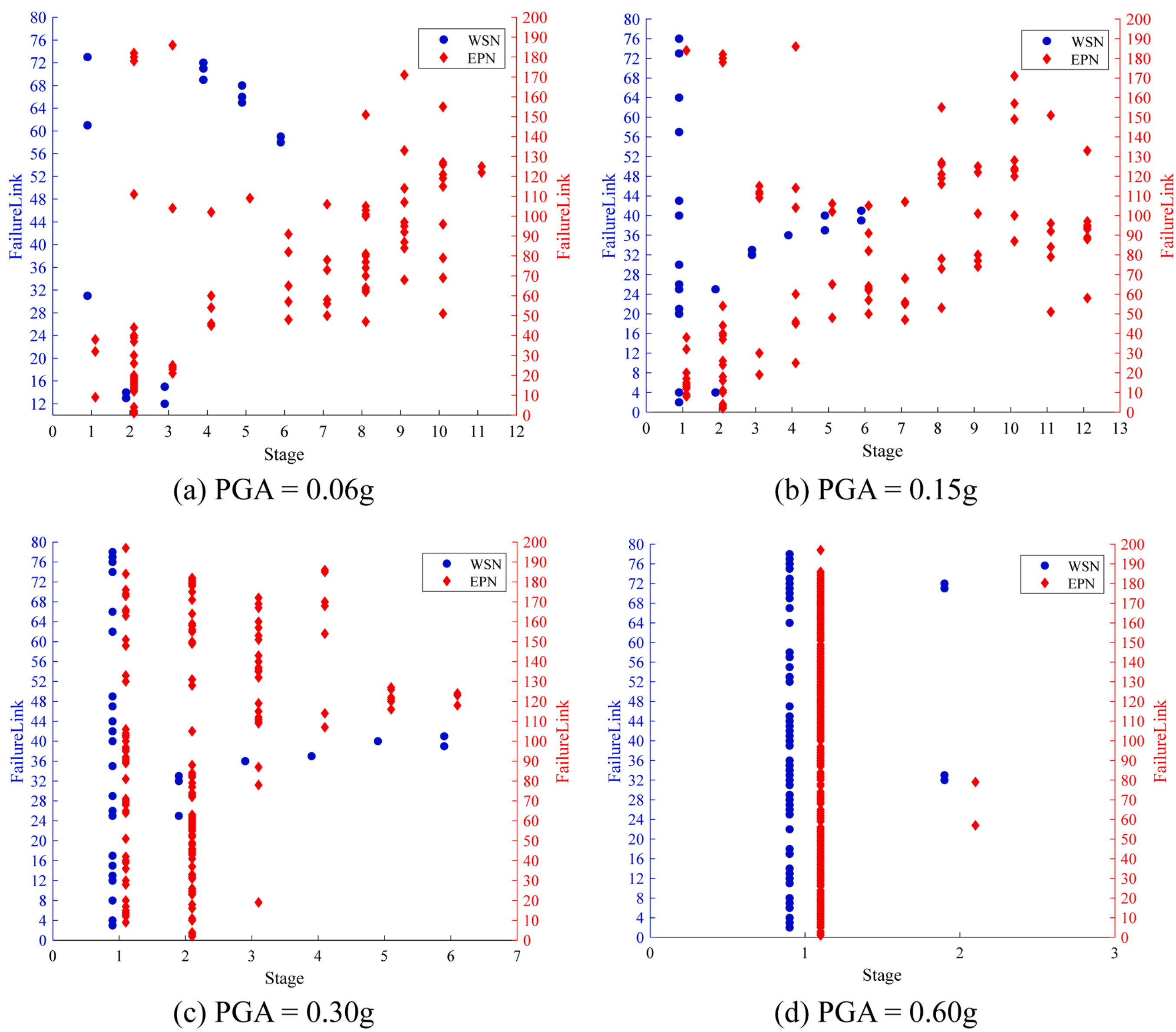
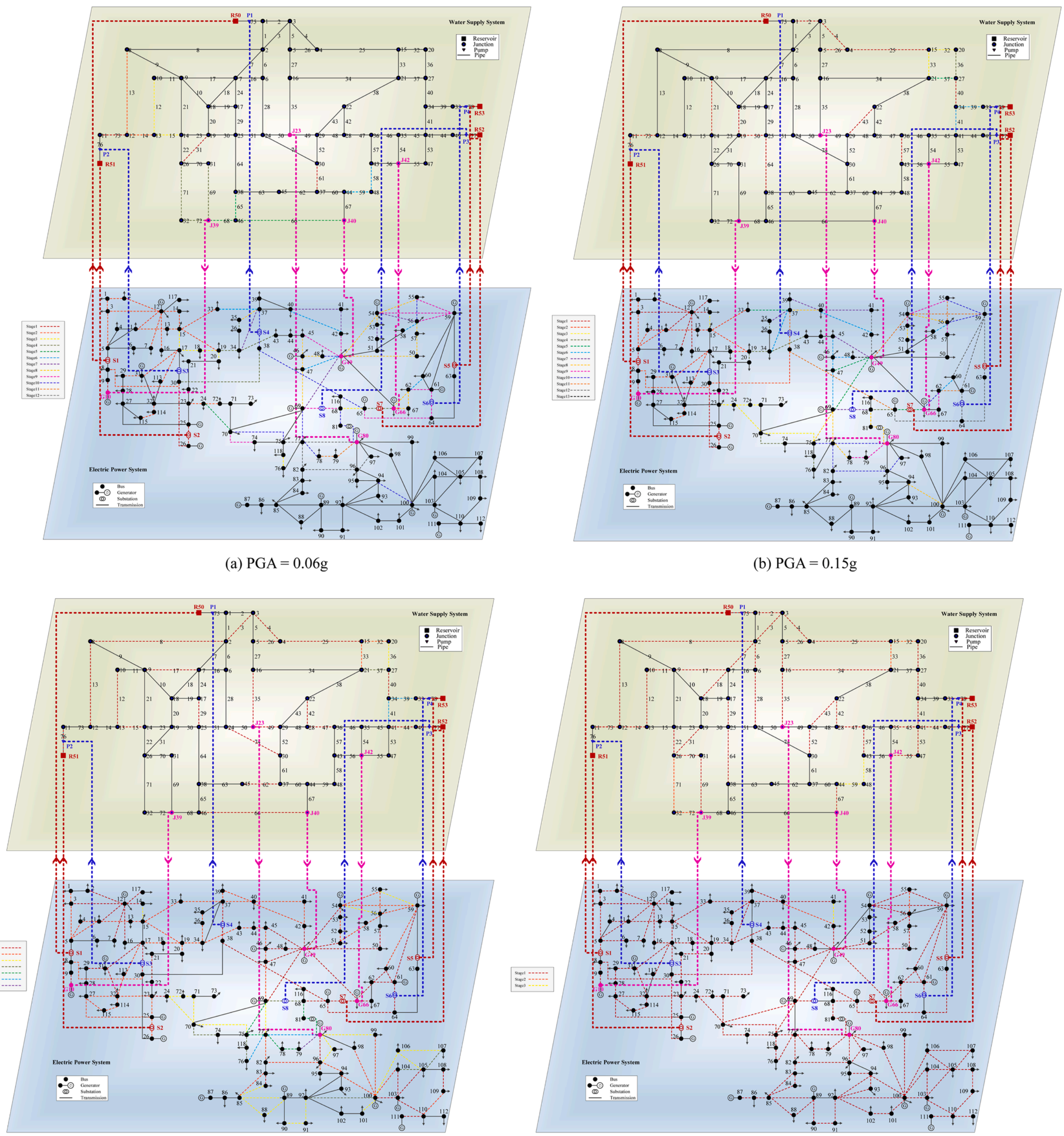


Fig. 4. Failure propagation results under different PGA.

transmission flow of the line, and the cascading failure propagation process was determined through the improved MATCASC model and the line capacity function. Under the above four seismic scenarios, the cascading failure simulations of the coupled WSN and EPN were implemented by the Monte Carlo method, with a number of 1000. In order to illustrate the representative results, this paper selected the results with the most failure propagation stages among 1000 simulations for visualization, as shown in Fig. 4. The propagation stages of cascading failure in the WSN were 6, 6, 6, and 3, respectively; while in the EPN they were 12, 13, 7, and 3, respectively. For the first three earthquake scenarios, there were relatively more cascading failure propagation stages in the WSN and EPN. The reason is that with the increase of PGA, the number of initial triggering components of cascading failure gradually increased. When PGA=0.6g, failure propagation stages in the two networks decreased. The main reason is that the initial damage caused by earthquakes to components is catastrophic, which may lead to large-scale system failure or collapse, and cascading effect transmissions dropped sharply. Fig. 5 shows the spatial distribution results of cascade failures under four earthquake scenarios. Among them, when PGA=0.06 and 0.15g, the spatial distribution of failure pipelines in the WSN was relatively scattered, and the failure transmission lines in the EPN were mainly distributed in the upper part of the network diagram. When



(a) PGA = 0.06g (b) PGA = 0.15g

(c) PGA = 0.30g (d) PGA = 0.60g

Fig. 5. Space distribution diagram of cascading failure under different PGA.

PGA=0.3 and 0.6g, the failure links in the WSN and EPN occupied almost the entire network, which indicated that the network functional loss was very serious. Overall, there were significantly more failed links in the EPN than in the WSN, showing that the EPN was much more damaged than the WSN.

3.3.2. Functional Loss Calculation

Table 3 summarizes the comparative results in functional loss of the WSN and EPN with and without functional coupling relationships. From a vertical perspective, as the PGA increased, the functional loss of the WSN and EPN with or without functional coupling relationships gradually increased. Taking the ENP as an example, when PGA=0.06g, the supply loss rates of the coupled and uncoupled EPN were 0.478 and 0.072, respectively, and the link loss rates were 0.443 and 0.073, respectively. When PGA increased to 0.6g, the supply loss rate and link loss rate both increased to 1.000 and 0.953, respectively. From a horizontal perspective, the functional loss of the coupled WSN and EPN was higher than that of the uncoupled WSN and EPN, and that of the EPN was more obvious. For instance, when PGA=0.15g, the supply loss rate and link loss rate of the coupled WSN were 0.085 and 0.152, respectively, which were greater than that of the uncoupled WSN. Results were similar for the EPN. It can be inferred that the coupled WSN and EPN suffered greater losses under earthquake scenarios, and the cascading failure range spread wider. In addition, as the PGA increased, the difference in functional loss between the WSN and EPN with or without functional coupling relationships gradually decreased. The major reason is that the initial damage caused by the earthquake to components gradually becomes serious and the functional loss gradually increases. In this circumstance, regardless of whether there is a functional coupling relationship, the cascading failure propagation effect in the network is not obvious.

3.4. Sensitivity Analysis

To further analyze the impact of functional coupling strength and tolerance parameter on cascading failure of the WSN and EPN, this paper conducted a sensitivity analysis of functional losses of the WSN and EPN to functional coupling strength and tolerance under four different earthquake scenarios. When conducting a sensitivity analysis of functional coupling strength, it is assumed that the functional coupling strength between the reservoir and the substation ranges from 0 to 1, with a step size of 0.02. The functional coupling strength between the pumping station and the substation is 1. The tolerance of each demand node is 0.5. In the EPN, the functional coupling strength between the power plant and the water demand node is set to 0~1, with a step size of 0.02. The tolerance of each transmission line is 0.2. When implementing sensitivity analysis of tolerance, the tolerance value range of each water demand node in the WSN is set to 0~1, with a step size of 0.02. The functional coupling strength between the reservoir and the substation is 0.5, and that of the pumping station and the substation is 1. In the EPN, the tolerance of each transmission line ranges from 0 to 1, with a step size of 0.02, while the functional coupling strength between the power plant and the water demand node is 0.5.

3.4.1. Sensitivity Analysis of Functional Loss to Functional Coupling Strength

Fig. 6 shows the sensitivity change trend of functional loss of the coupled WSN and EPN to functional coupling strength under four different earthquake scenarios. Overall, the functional coupling strength between the reservoir and the substation had little impact on the functional loss of the coupled WSN. A more obvious impact on the functional loss of the coupled EPN originated from the functional coupling strength between the power plant and the water demand node. As the PGA increased, the impact of functional coupling strength on the EPN functional loss gradually reduced. Specifically, as shown in Fig. 6 (a), with the increase of PGA, both the supply loss rate and the link loss rate of the WSN gradually increased. The change in functional coupling strength had almost no obvious impact on the link loss rate, but affected the supply loss rate slightly. As for the EPN, as shown in Fig. 6 (b), as the PGA increased, the supply loss rate and link loss rate also gradually increased. In addition, when PGA=0.06g, as the functional coupling strength between the power plant and the water demand node increased, the supply loss rate and link loss rate gradually increased, indicating that the functional loss was highly sensitive to changes in the functional coupling strength. When the PGA was greater than or equal to 0.15g, the supply loss rate and link loss rate changed slightly, suggesting that the sensitivity of functional loss to the change of functional coupling strength gradually reduced. The main reason is that the initial damage of the earthquake to the coupled EPN could be relatively serious, while the functional coupling strength has little effect on the cascading failure of the EPN, resulting in a decreased sensitivity to functional loss.

3.4.2. Sensitivity Analysis of Functional Loss to Tolerance

Regarding the sensitivity analysis of functional loss to tolerance, numerical simulations of the functional loss changes of the WSN and EPN with and without functional coupling relationships under four different earthquake scenarios were implemented. In the numerical simulation without functional coupling relationships, except for the functional coupling strength, other parameters remain consistent with the settings with functional coupling relationships. Fig. 7 exhibits the sensitivity change results of the functional loss of the WSN and EPN with or without functional coupling relationships to the tolerance. The dashed line represents the results of the coupled networks, and the solid line displays that of the uncoupled networks. Overall, the functional losses of the coupled WSN and EPN were more serious than those of the uncoupled networks. Specifically, as shown in Fig. 7 (a), with the increase of PGA, both the link loss rate and supply loss rate of the WSN gradually increased; while the tolerance of each water demand node had almost no impact on the link loss rate, but had a slight impact on the supply loss rate. In Fig. 7 (b), as the PGA increased, the link loss rate and supply loss rate of the coupled and uncoupled EPN also gradually increased. In addition, when PGA=0.06g, 0.15g, and 0.3g, as the

Table 3
Comparison in functional loss of the WSN and EPN with or without functional coupling relationships.

	Metric							
	EPN				WSN			
	F_{loss}^{DSR}		F_{loss}^{LSR}		F_{loss}^{DSR}		F_{loss}^{LSR}	
PGA/g	FC	NFC	FC	NFC	FC	NFC	FC	NFC
0.06	0.478	0.072	0.443	0.073	0.083	0.076	0.049	0.048
0.15	0.721	0.678	0.650	0.637	0.085	0.081	0.152	0.148
0.3	0.932	0.919	0.774	0.772	0.222	0.192	0.332	0.328
0.6	1.000	1.000	0.953	0.953	0.327	0.315	0.671	0.670

Note: FC denotes that there are functional coupling relationships between the EPN and WSN; NFC denotes that there is no functional coupling relationship between the EPN and WSN.

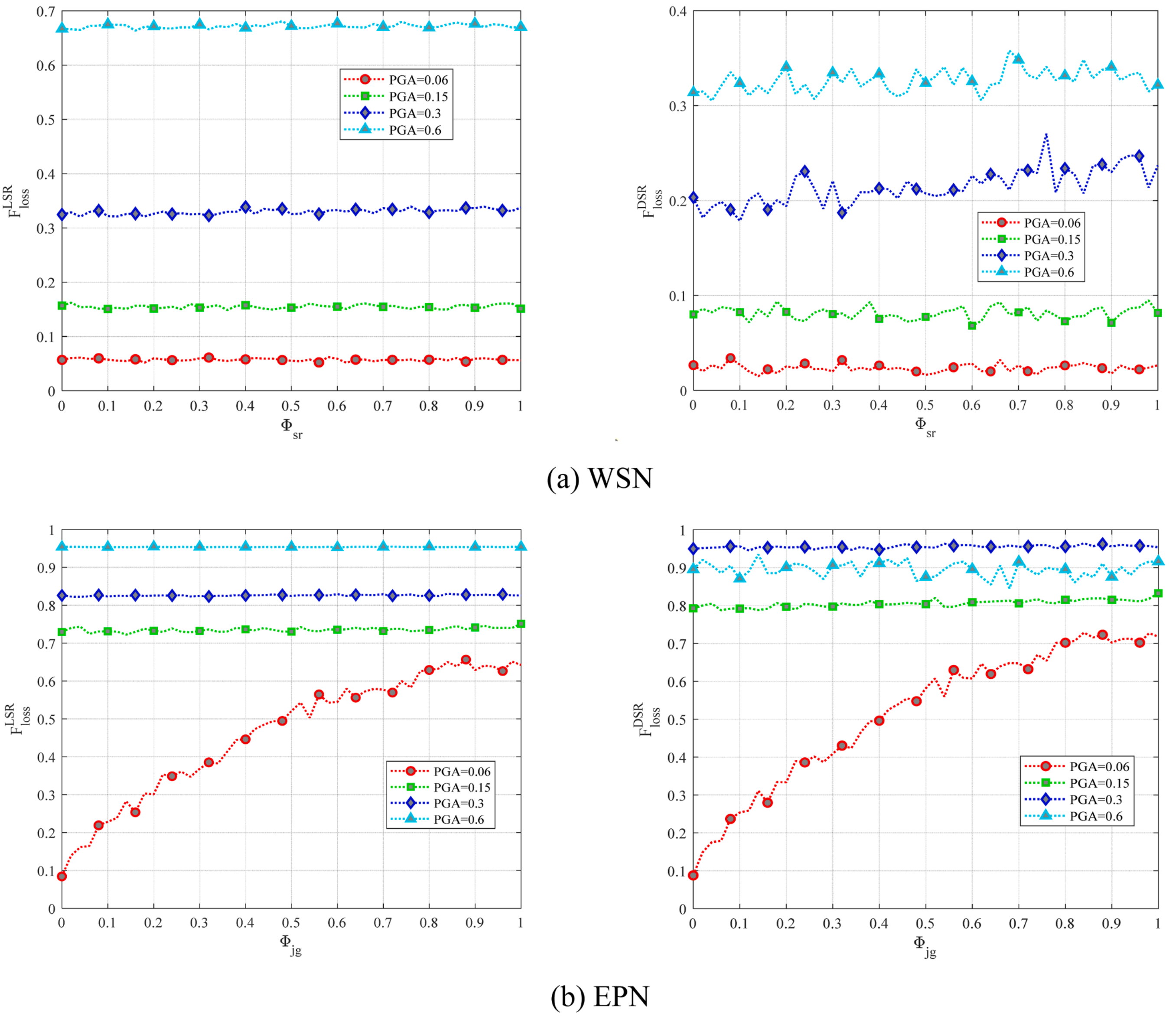


Fig. 6. Sensitivity Analysis of Functional Loss to Functional Coupling Strength.

tolerance of the transmission line increased, the link loss rate and the supply loss rate of the coupled and uncoupled EPN gradually decreased. When PGA=0.6g, little effect on the functional loss of the coupled and uncoupled EPN originated from the tolerance of each transmission line, indicating that the sensitivity of functional loss to changes in the tolerance reduced. The major reason may be that the initial damage of the earthquake to the EPN is relatively serious, while the tolerance has little effect on the cascading failure of the EPN, resulting in a reduced sensitivity to functional loss. Further, when PGA=0.06g, the functional loss of the coupled EPN was significantly higher than that of the uncoupled. In the other three earthquake scenarios, there was little difference in the degree of functional loss between the coupled and uncoupled EPN.

4. Discussion

According to the cascading failure simulation results in Section 3.3, it can be reasonably concluded that the cascading failure propagation range of the coupled WSN and EPN is wider and the functional losses are more serious than that of the uncoupled networks. In addition, it can be seen from Fig. 4 that compared with the WSN, more cascading failure propagation stages and a larger failure propagation dimension occurred in the EPN under four different earthquake scenarios. It also can be observed from Table 3 that the supply loss rate and link loss rate of the coupled WSN and EPN were higher than that of the uncoupled networks, and as the PGA increased, the difference in functional loss between the coupled and uncoupled network gradually decreased. Based on the sensitivity analysis results in Section 3.4, it can be concluded that the functional coupling strength had almost no impact on the link loss rate of the WSN and affected the supply loss rate slightly. For the EPN, a significant impact on the functional loss came from the functional coupling strength when PGA = 0.06g. Under the other three earthquake scenarios, there is almost no impact on the functional loss. Wang et al pointed out that there were differences in component tolerance of infrastructure networks, depending on network characteristics, operating mechanisms, etc., and the impact on the propagation path of system cascading failures was also different [41]. This conclusion is consistent with the results of the sensitivity analysis of functional loss to different system tolerances in this paper. It can be seen from Fig. 7 that the node tolerance had almost no impact on the link loss rate of the WSN and a small impact on the supply loss rate, indicating that the functional loss was weakly sensitive to node tolerance. The transmission line tolerance had a significant impact on the link loss rate and supply loss

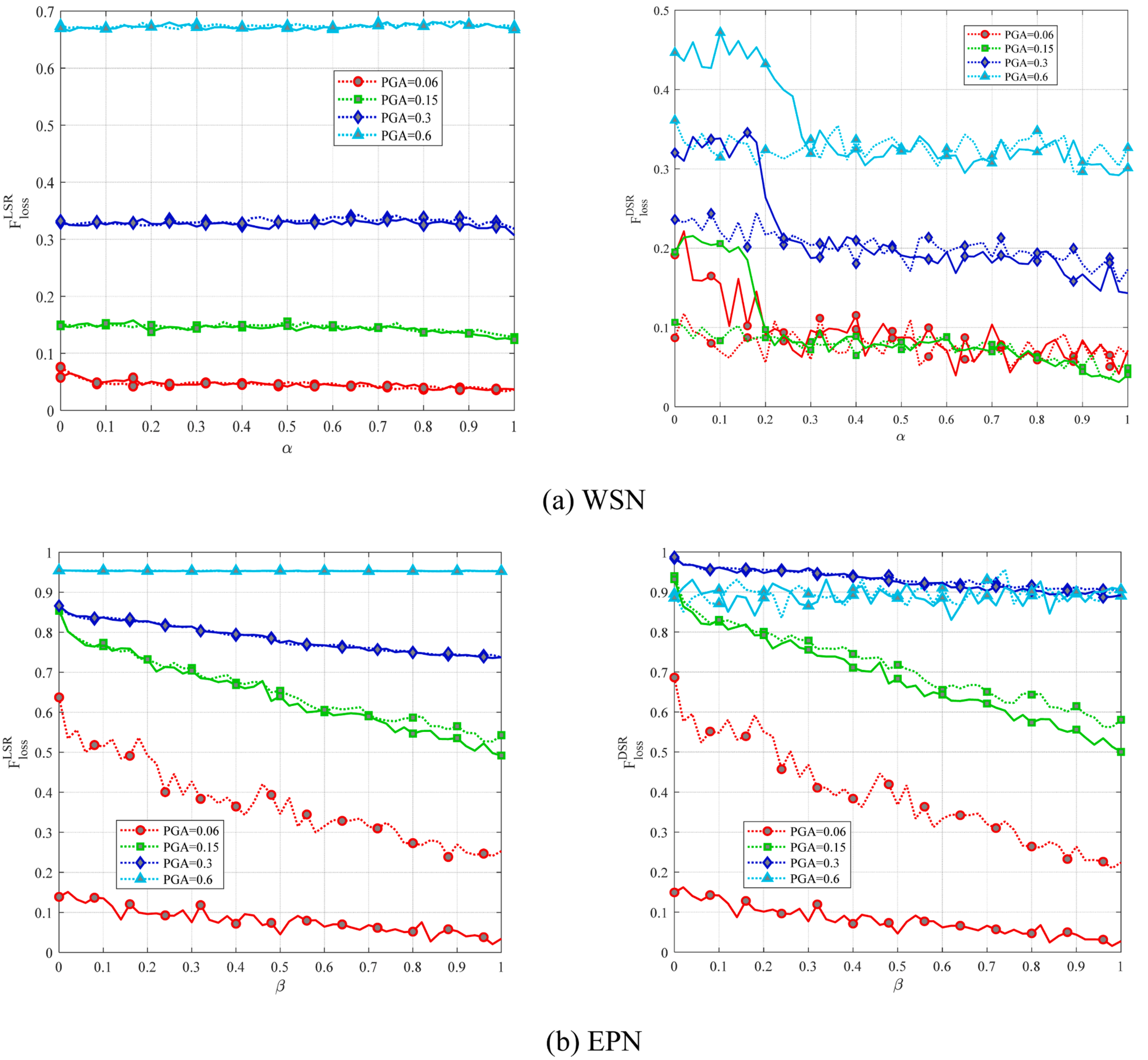


Fig. 7. Sensitivity Analysis of Functional Loss to Tolerance.

rate of the EPN, suggesting that functional loss was highly sensitive to it.

In conclusion, it can be inferred that the impact of functional coupling strength and component tolerance on the cascading failure and functional loss of the IWPN cannot be ignored. Otherwise, the evaluation results of post-earthquake network functional loss will be seriously underestimated. The results of this study can improve understanding of cascading failure propagation modeling methods in the WSN and EPN that accurately consider the functional coupling relationships between the two networks. It is worth noting that the proposed method can provide a theoretical reference for identifying and considering system functional coupling relationships in the process of network topology modeling and cascading failure modeling of interdependent infrastructure systems. In addition, a better understanding of the interdependencies between the WSN and EPN can be obtained for system managers, operations, and emergency personnel, which will help accurately estimate the impact of earthquakes on the IWPN. It is conducive to establishing a better decision support system for the design, construction, and maintenance of the WSN and EPN, and improving the system's prevention and response capabilities to disasters.

5. Conclusion

As the urban typical infrastructure system, the WSN and EPN ensure the stable development of the urban socio-economy and people's production and life. Under earthquakes, minor system localized damage may cause large-scale system failure propagation due to the interdependence between the two networks. Moreover, it may also bring disastrous consequences to the city. Therefore, this paper conducts cascading failure analysis for the IWPN under earthquake scenarios. First, from the perspective of the functional supply and demand relationship of system components, this paper defined the functional coupling relationships between the WSN and EPN and established the topological structure of the IWPN based on complex network methods. Subsequently, the fragility analysis was used to determine the probability of different damage states of components. The joint probability and functional coupling strength were introduced to calculate the failure probability of coupled components. And then, the node load function and line capacity function were introduced as the judgment conditions for cascading failure of water network nodes and power network lines, respectively. The cascading failure process of the WSN and EPN was conducted based on the dynamical flow model. Further, a calculation method for the functional loss of the WSN and EPN was proposed. Finally, taking the coupling WSN of a certain city and IEEE 118-node network as an example, the Monte Carlo method was used to simulate the cascading failure process of the WSN and EPN with and without functional coupling relationships under four different earthquake scenarios. Moreover, sensitivity analysis was conducted on the functional coupling strength and tolerance threshold to obtain the functional loss impact of WSN and EPN. The results showed that cascading failures in the IWPN spread wider and caused more serious functional losses. In addition, under four earthquake scenarios, the functional loss of the WSN was weakly sensitive to the functional coupling strength and node load. However, the sensitivity of the functional loss of the EPN to functional coupling strength and link capacity was observable. But as the PGA increased, the sensitivity was gradually weakened.

Universally, the proposed cascading failure analysis method can be applied to other infrastructure networks. Taking the IWPN as the research objects, complex network methods are used to construct the topological model and the functional coupling relationships from the perspective of the functional supply and demand relationship of system components are considered. Therefore, for different infrastructure networks, the mutual coupling relationships between system components and physical connections as well as their functions should be further analyzed. In addition, the tolerance of the WSN and EPN in this research is a set value. For one thing, existing research has analyzed its resistance to cascading failures in the WSN and EPN. For another, the main purpose of this article is to reveal the cascading failure rules of the post-earthquake IWPN considering physical operational characteristics, analyze the impact of functional coupling relationships on the propagation of cascading failures in the WSN and EPN, and further quantify the functional loss of the two networks. Therefore, the tolerance value can be further determined based on the physical operation conditions of the specific infrastructure system.

From the perspective of functional coupling relationship and disaster spread, the cascading failure analysis and functional loss quantification of the WSN and EPN under earthquakes are carried out. It can provide support for cascading failure modeling research on multi-layer interdependent infrastructure systems, and provide certain references for formulating disaster prevention and reduction measures and seismic performance improvement strategies for interdependent infrastructure networks. In future research, it can be expanded from the following aspects: 1) carry out cascading failure analysis on the WSN and EPN of different sizes, conduct comparative analysis on different coupled networks under earthquakes of the same magnitude, and propose targeted seismic performance improvement strategies. 2) expand double-layer infrastructure networks into multi-layer infrastructure networks, including transportation networks, gas networks, etc., to handle cascading failures of multi-layer interdependent infrastructure networks, and provide support for improving the disaster prevention and reduction capabilities of urban infrastructure networks.

CRediT authorship contribution statement

Yang Li: Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Mingyuan Zhang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Key R&D Program of China (No. 2023YFC3805202) and the National Natural Science Foundation of China (No. 52278485).

Data availability

Data will be made available on request.

References

[1] Fan X, Zhang L, Wang J, Ren Y, Liu A. Analysis of faulting destruction and water supply pipeline damage from the first mainshock of the February 6, 2023 Türkiye earthquake doublet. Earthq Sci 2024;37:78–90. https://doi.org/10.1016/j.eqs.2023.11.004.

[2] Buldyrev SV, Parshani R, Paul G, Stanley HE, Havlin S. Catastrophic cascade of failures in interdependent networks. Nature 2010;464:1025–8. https://doi.org/10.1038/nature08932.

[3] Zhou J, Coit DW, Felder FA, Tsianikas S. Combined optimization of system reliability improvement and resilience with mixed cascading failures in dependent network systems. Reliab Eng Syst Saf 2023;237:109376. https://doi.org/10.1016/j.ress.2023.109376.

[4] Rinaldi SM, Peerenboom JP, Kelly TK. Identifying, understanding, and analyzing critical infrastructure interdependencies. IEEE Control Syst Mag 2001;21:11–25. https://doi.org/10.1109/37.969131.

[5] Ouyang M. Review on modeling and simulation of interdependent critical infrastructure systems. Reliab Eng Syst Saf 2014;121:43–60. https://doi.org/10.1016/j.ress.2013.06.040.

[6] Utne IB, Hokstad P, Vatn J. A method for risk modeling of interdependencies in critical infrastructures. Reliab Eng Syst Saf 2011;96:671–8. https://doi.org/10.1016/j.ress.2010.12.006.

[7] Franchina L, Carbonelli M, Gratta L, Crisci M, Perucchini D. An impact-based approach for the analysis of cascading effects in critical infrastructure. Int J Crit Infrastructures 2011;7:73–90.

[8] Santos JR, Haimes YY, Lian C. A framework for linking cybersecurity metrics to the modeling of macroeconomic interdependencies. Risk Anal 2007;27:1283–97. https://doi.org/10.1111/j.1539-6924.2007.00957.x.

[9] Haimesl YY, Jiang P. Leontief-based model of risk in complex interconnected infrastructures. J Infrastruct Syst 2001;7:1–12.

[10] Basu N, Pryor R, Quint T. ASPEN : A microsimulation model of the economy. Comput Econ 12 1998;12:223–41.

[11] Approach S, Mahmood A, Montagna F. Making lean smart by using system-of-systems' approach. IEEE Syst J 2013;7:537–48. https://doi.org/10.1109/JSYST.2013.2244801.

[12] Johansson J, Hassel H. An approach for modelling interdependent infrastructures in the context of vulnerability analysis. Reliab Eng Syst Saf 2010;95:1335–44. https://doi.org/10.1016/j.ress.2010.06.010.

[13] Ii EEL, Mitchell JE, Wallace WA. Restoration of services in interdependent infrastructure systems : A network flows approach. IEEE Trans Syst Man, Cybern Part C Appl Rev 2007;37:1303–17.

[14] Sun W, Bocchini P, Davison BD. Overview of interdependency models of critical infrastructure for resilience assessment. Nat Hazards Rev 2022;23:1–14. https://doi.org/10.1061/(ASCE)NH.1527-6996.0000535.

[15] Mitsova D. Integrative interdisciplinary approaches to critical infrastructure interdependency analysis. Risk Anal 2021;41:1111–7. https://doi.org/10.1111/risa.13129.

[16] Sharma N, Nocera F, Gardoni P. Classification and mathematical modeling of infrastructure interdependencies. Sustain Resilient Infrastruct 2021;6:4–25. https://doi.org/10.1080/23789689.2020.1753401.

[17] Holden R, Val D V, Burkhard R, Nodwell S. A network flow model for interdependent infrastructures at the local scale. Saf Sci 2013;53:51–60. https://doi.org/10.1016/j.ssci.2012.08.013.

[18] Valdez LD, Shekhtman L, La Rocca CE, Zhang X, Buldyrev SV, Trunfio PA, et al. Cascading failures in complex networks. J Complex Netw 2020;8:cnaa013. https://doi.org/10.1093/comnet/cnaa013.

[19] Rahnamay-naeini M, Hayat MM. Cascading failures in interdependent infrastructures : an interdependent Markov-chain approach. IEEE Trans Smart Grid 2016;7:1997–2006. https://doi.org/10.1109/TSG.2016.2539823.

[20] Wu Y, Chen Z, Zhao X, Gong H, Su X, Chen Y. Propagation model of cascading failure based on discrete dynamical system. Reliab Eng Syst Saf 2021;209:107424. https://doi.org/10.1016/j.ress.2020.107424.

[21] Guidotti R, Chmielewski H, Unnikrishnan V, Gardoni P, Mcallister T, De Lindt J Van, et al. Modeling the resilience of critical infrastructure : the role of network dependencies. Sustain Resilient Infrastruct 2016;9689:1–16. https://doi.org/10.1080/23789689.2016.1254999.

[22] Xiao Y, Zhao X, Wu Y, Chen Z, Gong H. Seismic resilience assessment of urban interdependent lifeline networks. Reliab Eng Syst Saf 2022;218:108164. https://doi.org/10.1016/j.ress.2021.108164.

[23] Oikonomou K, Parvania M, Member S. Optimal coordinated operation of interdependent power and water distribution systems. IEEE Trans Smart Grid 2020;11:4784–94.

[24] Pournaras E, Taormina R, Galelli S, Kooij R. Cascading failures in interconnected power-to-water networks. Perform Eval Rev 2020;47:16–20.
[25] Zhang Y, Yang N, Lall U. Modeling and simulation of the vulnerability of interdependent power-water infrastructure networks to cascading failures. J Syst Sci Syst Eng 2016;25:102–18. https://doi.org/10.1007/s11518-016-5295-3.
[26] Lee S, Choi M, Lee HS, Park M. Bayesian network-based seismic damage estimation for power and potable water supply systems. Reliab Eng Syst Saf 2020;197:106796. https://doi.org/10.1016/j.ress.2020.106796.
[27] Hou B, Huang J, Miao H, Zhao X, Wu S. Seismic resilience evaluation of water distribution systems considering hydraulic and water quality performance. Int J Disaster Risk Reduct 2023;93:103756. https://doi.org/10.1016/j.ijdrr.2023.103756.
[28] Liu X, Xie Q, Liang H, Zhang X. Post-earthquake recover strategy for substations based on seismic resilience evaluation. Eng Struct 2023;279:115583. https://doi.org/10.1016/j.engstruct.2022.115583.
[29] Federal emergency management agency. Hazus earthquake model technical manual. Washington DC: 2022.
[30] Guidotti R, Gardoni P, Rosenheim N. Integration of physical infrastructure and social systems in communities' reliability and resilience analysis. Reliab Eng Syst Saf 2019;185:476–92. https://doi.org/10.1016/j.ress.2019.01.008.
[31] Zhang P, Peeta S. A generalized modeling framework to analyze interdependencies among infrastructure systems. Transp Res Part B Methodol 2011;45:553–79. https://doi.org/10.1016/j.trb.2010.10.001.
[32] General administration of quality supervision I and Q of the PR of C. Seismic ground motion parameters zonation map of china. Standardization Administration of the People's Republic of China; 2015.
[33] Guideline for evaluation of seismic resilience assessment of urban engineering systems. Research Institute of Standards and Norms Ministry of Housing and Urban-Rural Development; 2022.
[34] Zhao XD, Chen ZL, Xu JH, Tang HZ. Seismic vulnerability of urban double-layer interdependent lifeline network. J Zhejiang Univ (Eng Sci) 2020;54:767–77. https://doi.org/10.3785/j.issn.1008-973X.2020.04.016.
[35] Wu J, Dueñas-Osorio L. Calibration and validation of a seismic damage propagation model for interdependent infrastructure systems. Earthq Spectra 2013;29:1021–41. https://doi.org/10.1193/1.4000160.
[36] Yoon S, Lee YJ, Jung HJ. Flow-based optimal system design of urban water transmission network under seismic conditions. Water Resour Manag 2020;34:1971–90. https://doi.org/10.1007/s11269-020-02541-4.
[37] Wagner JM, Shamir U, Marks DH. Water distribution reliability: simulation methods. J Water Resour Plan Manag 1988;114:276–94. https://doi.org/10.1061/(asce)0733-9496(1988)114:3(276).
[38] Han Z, Ma D, Hou B, Wang W. Post-earthquake hydraulic analyses of urban water supply network based on pressure drive demand model. Sci Sin Technol 2019;49:351–62. https://doi.org/10.1360/N092017-00429.
[39] Shuang Q, Zhang M, Yuan Y. Node vulnerability of water distribution networks under cascading failures. Reliab Eng Syst Saf 2014;124:132–41. https://doi.org/10.1016/j.ress.2013.12.002.
[40] Standardization administration of the People's republic of China. Chin Seismic Intens scale. 2020.
[41] Wang F, Magoua JJ, Li N, Fang D. Assessing the impact of systemic heterogeneity on failure propagation across interdependent critical infrastructure systems. Int J Disaster Risk Reduct 2020;50:101818. https://doi.org/10.1016/j.ijdrr.2020.101818.